Elementary Particles

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Introduction

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1 . Overview to Particle Physics

1.1 Elementary particles and interactions: the energy-scale viewpoint

The basic questions of particle physics are deceptively simple:

- ◇ What are the elementary building blocks of Nature?
- ◇ What are the interactions among them?

A key lesson is that the answer depends on *the energy scale* at which we probe Nature: as we increase the typical energy (or momentum transfer) of our experiments, new degrees of freedom become relevant and the effective description changes.

A pragmatic (and historically successful) way to organize our knowledge is to relate *energy scales* to the *particles we can resolve*, the *interactions we observe*, their *ranges*, and their *mediators*. At very low energies we describe matter in terms of atoms and effective potentials; at higher energies we resolve nuclei, then nucleons, and eventually quarks, leptons and gauge bosons.

1.1.1 A qualitative "energy ladder"

Scale	Degrees of freedom	Interactions	Range
$\lesssim$ eV	atoms (effective)	effective (atomic/molecular)	long
$\sim$ keV	e^-, nuclei	e.m.	long
$\sim$ MeV	$e^\pm$, p, n, ν_e	e.m. + nuclear + weak	long / short / contact
$\sim$ GeV	$\mu^\pm$, ν_μ, hadrons	e.m. + strong + weak	long / short / contact
$\sim$ 10 GeV	$\tau^\pm$, ν_τ, u, d, s, c, b	e.m. + QCD + weak	long / short / contact
$\sim$ TeV	$W^\pm$, Z^0, t, H	e.m. + QCD + EW	long / short / short

1.2 Natural units and physical intuition

At subatomic scales we inevitably need quantum mechanics, hence $\hbar$ appears in formulas. At high velocities we need special relativity, hence c appears as well. To avoid a constant

proliferation of $\hbar$ and c, we adopt *natural units*

$$\hbar = c = 1, \tag{1.1}$$

so that any dimensionful quantity can be expressed as a power of energy. In this convention, masses, momenta and energies share the same units (eV, MeV, GeV, ...).

To recover SI units, one reinserts appropriate powers of $\hbar$ and c and uses 1 eV $\simeq 1.6 \times 10^{-19}$ J. Useful reference scales include

- $E_{\text{Rydberg}} = 13.6$ eV,
- $m_e \simeq 0.5$ MeV,
- $m_p \simeq 940$ MeV.

For distances and cross sections, it is common to use

- the fermi (fm): 1 fm $= 10^{-15}$ m, and $\hbar c \simeq 197$ MeV fm,
- the barn (b): 1 b $= 100$ fm^2.

1.3 The Standard Model and what lies beyond

The Standard Model (SM) describes elementary particles and their interactions up to energies of order $\mathcal{O}(10 \text{ TeV})$. It contains:

- **Matter fields** (spin $J = \frac{1}{2}$): quarks and leptons. Charged leptons and quarks are massive; neutrino masses are very small.
- **Gauge bosons** (spin $J = 1$): photon and gluons are massless; $W^{\pm}$ and Z^0 are massive.
- **Higgs boson** (spin $J = 0$), massive.

Despite its success, there are clear indications that the SM is not the final story:

- **Gravity:** extremely weak at subatomic distances but long-range, suggesting a massless mediator (graviton) in a quantum description.
- **Dark matter:** gravitational evidence at galactic/cosmological scales with no electromagnetic signals, suggesting at least one new particle.

1.4 Energy frontiers in Nature

The highest man-made energies are achieved in colliders. At present, the LHC reaches ~ 13 TeV in the center-of-mass energy. Future proposals include FCC(hh) at ~ 100 TeV and lepton/hadron options at intermediate energies.

Nature also provides *cosmic rays* with energies far beyond colliders: ultra-high energy cosmic rays satisfy $E > 10^4$ TeV, with record events around $E \sim 3 \times 10^8$ TeV (Utah, 1991).

1.5 QFT as the language of particle physics

Combining quantum mechanics with special relativity naturally leads to *quantum field theory* (QFT):

- states can contain an *arbitrary number of particles* (Fock space),
- elementary particles correspond to *unitary irreducible representations* of the Poincaré group $\mathrm{ISO}(3,1)$.

The irreducible unitary representations are characterized by

$$m^2 \in \mathbb{R}_+, \qquad \mathbf{p} \in \mathbb{R}^3, \qquad J \in \tfrac{1}{2}\mathbb{Z}_+, \qquad J_3 \in \{-J, -J+1, \ldots, J\}, \tag{1.2}$$

with the special feature that for massless particles only the extreme spin projections appear.

1.6 Strength of interactions and unification scales

A convenient way to compare interactions is through dimensionless couplings (or effective dimensionless combinations at a given scale). At $E \sim m_p \sim 1$ GeV (low energies) and $E \sim m_Z \sim 100$ GeV (high energies), one finds qualitatively:

- **Strong:** $\alpha_s(m_p) \sim 1$ but $\alpha_s(m_Z) \sim 0.1$ (running).
- **Electromagnetic:** $\alpha(m_p) \simeq 1/137$, $\alpha(m_Z) \simeq 1/129$.
- **Weak:** at low energies appears through $G_F m_p^2 \sim 10^{-5}$, while at m_Z one may define $\alpha_W(m_Z) = \dfrac{m_W^2 G_F}{4\pi} \sim \mathcal{O}(1)$.
- **Gravity:** $G m_p^2 \sim 10^{-39}$, negligible in particle physics.

Two important conceptual scales follow from extrapolations:

- **Electroweak unification:** around $E \sim m_Z$, electromagnetic and weak strengths become comparable.
- **Grand unification:** strong and electroweak strengths may meet around $E \sim 10^{16}$ GeV.
- **Planck scale:** gravity becomes comparable to other interactions at $E \sim G^{-1/2} \sim 10^{19}$ GeV.

1.7 Conservation laws and discrete symmetries

Conservation laws are deeply tied to symmetries:

- **Space-time symmetries:** energy–momentum conservation; angular momentum conservation.
- **Internal symmetries:** electric charge; baryon number B; lepton number L.

In the SM, individual lepton flavor numbers (electron, muon, tau) are conserved at the Lagrangian level, but neutrino oscillations show this is not an exact symmetry in Nature. Quantum effects also violate B and L separately at an unobservably small level, leaving $B+L$ as the truly conserved combination within the SM description at accessible energies.

Strong and electromagnetic interactions conserve additional discrete symmetries: parity (P), charge conjugation (C), time reversal (T), and (approximately) flavor.

1.8 Hadrons: mesons, baryons and color

Although quarks and gluons are the building blocks of the strong interaction in the SM, in Nature we observe only *hadrons*: bound states that are color singlets. The simplest hadrons are:

- **Mesons:** quark–antiquark ($q\bar{q}$),
- **Baryons:** three quarks (qqq).

1.8.1 Isospin and light mesons/baryons

Many light hadrons can be organized into representations of SU(2) isospin. For example, pions form an isotriplet (π^+, π^0, π^-) and nucleons form an isodoublet (p, n). A simple quark-model picture for light mesons treats them as $q\bar{q}$ with

$$J = \tfrac{1}{2} \otimes \tfrac{1}{2} = 0 \oplus 1, \qquad I = \tfrac{1}{2} \otimes \tfrac{1}{2} = 0 \oplus 1, \tag{1.3}$$

with u and d quarks carrying charges $Q_u = \frac{2}{3}e$ and $Q_d = -\frac{1}{3}e$.

Similarly, light baryons can be described as three-quark states with vanishing orbital angular momentum:

$$\tfrac{1}{2} \otimes \tfrac{1}{2} \otimes \tfrac{1}{2} = \tfrac{1}{2} \oplus \tfrac{1}{2} \oplus \tfrac{3}{2}, \tag{1.4}$$

leading to, for instance, the nucleon doublet and the Δ quartet.

1.8.2 The Pauli principle and the need for color

A puzzle arises because the spin-3/2 Δ states are totally symmetric under exchange of quarks, apparently conflicting with the Pauli principle. The resolution is the introduction of a new quantum number: **color**. Quarks transform in the fundamental representation $\mathbf{3}$ of $\mathrm{SU}(3)_{\mathrm{color}}$, and physical states must be color singlets.

Group-theory products show why mesons and baryons can be colorless, while isolated quarks and gluons cannot appear as asymptotic states:

$$\mathbf{3} \otimes \mathbf{3}^* = \mathbf{1} \oplus \mathbf{8}, \qquad \mathbf{3} \otimes \mathbf{3} = \mathbf{3}^* \oplus \mathbf{6}, \qquad \mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{10}. \tag{1.5}$$

Hence $q\bar{q}$ and qqq can contain singlets, but single quarks, gluons (octet), or qq states cannot be observed as free particles.

1.8.3 Exotic hadrons

QCD does not forbid hadrons beyond mesons and baryons: tetraquarks and pentaquarks are allowed. Their unambiguous identification is easier in systems with heavy quarks, where the spectrum is cleaner and thresholds are better controlled.

At low energies, the strong interaction is effectively short-range because the lightest physical excitations exchanged between hadrons are pions, with $m_\pi \sim 140$ MeV.

1.9 Weak interactions and flavor quantum numbers

Hadron resonances such as ρ, ω or Δ have typical decay widths $\Gamma \sim 1$–100 MeV, characteristic of strong decays. However, the lightest isospin multiplets $(\pi^\pm, \pi^0, p, n)$ have much smaller widths:

- π^0 decays electromagnetically with $\Gamma \sim 10^{-5}$ MeV,
- $\pi^\pm$ and n decay weakly with Γ as small as 10^{-14}–10^{-24} MeV.

Weak interactions are also needed to describe muon decay and strange-hadron decays.

1.9.1 Strangeness

Strange hadrons are produced in strong interactions as particle–antiparticle pairs but decay via the weak interaction. This motivates an approximately conserved quantum number in strong processes: **strangeness** S. Strange mesons and baryons populate isospin multiplets, and the lightest ones decay weakly.

1.10 Discrete symmetry violation and CP

Parity (P) and charge conjugation (C) are maximally violated in weak interactions. A further, smaller violation of CP is observed in neutral kaon decays. This will play a central role in flavor physics and in understanding the origin of matter–antimatter asymmetry.

1.11 More generations: charm, bottom and top

The existence of the charm quark was predicted (around 1970) because flavor-changing neutral currents were not observed. Charm was discovered in 1974 through the resonance $e^+e^- \to \gamma^* \to J/\psi$, interpreted as a $c\bar{c}$ bound state.

Bottom was discovered in 1977 through $e^+e^- \to \gamma^* \to \Upsilon$, interpreted as a $b\bar{b}$ bound state. The electroweak theory then implied a third-generation partner, the top quark, and the bottom–top structure is crucial for understanding CP violation in kaon (and later B) systems.

The top quark was discovered in 1995 at the Tevatron ($p\bar{p}$ collider), with $m_t \simeq 172$ GeV and a large width $\Gamma_t \sim 1.4$ GeV. It decays weakly before hadronizing, making it unique

among quarks.

1.12 Summary and "big picture"

Particle physics is organized around:

- the energy-scale dependence of effective degrees of freedom,
- QFT as the consistent framework combining QM and SR,
- gauge symmetries and their associated forces,
- confinement and color singlets in QCD,
- weak interactions, flavor, and discrete symmetry violation.

These themes will guide the construction of concrete field theories and the analysis of real processes in the next chapters.

2 . Fields for Free Particles and Discrete Symmetries

2.1 Overview: from relativistic wave equations to QFT

We need a formalism that:

- ◇ combines Quantum Mechanics (QM) and Special Relativity (SR),
- ◇ describes processes with an arbitrary number of particles.

The outcome is *Quantum Field Theory* (QFT). A systematic way to build a QFT for free particles is:

1. Find a suitable (relativistic) wave equation.
2. Find a Lagrangian whose Euler–Lagrange equations reproduce it.
3. Apply canonical quantization rules.

In QFT, wave functions become *operators*, and the arbitrary coefficients in general solutions become *creation and annihilation operators*. We will go through:

- ◇ Schrödinger field (non-relativistic),
- ◇ scalar field (spin 0, Klein–Gordon),
- ◇ Dirac field (spin 1/2),
- ◇ vector field (spin 1, Proca/Maxwell),

and we will discuss the implementation of discrete symmetries P, C and T in each case.

2.2 The Schrödinger field (non-relativistic)

2.2.1 Equation of motion and Lagrangian

The Schrödinger equation is

$$\left(i\frac{\partial}{\partial t} + \frac{\nabla^2}{2m}\right)\psi(t,\mathbf{x}) = 0. \tag{2.1}$$

If the particle has spin s, each of the $2s+1$ components satisfies the same equation. This equation follows from the action $S = \int dt\, L$, with

$$L = \int d^3x\, \mathcal{L}, \qquad \mathcal{L}(t,\mathbf{x}) = \psi^\dagger(t,\mathbf{x}) \left(i\partial_0 + \frac{\nabla^2}{2m} \right) \psi(t,\mathbf{x}). \tag{2.2}$$

2.2.2 Canonical quantization and Fock space

Define the canonical momenta (for components $a = 1, \ldots, 2s+1$)

$$\Pi_a(x) \equiv \frac{\partial \mathcal{L}}{\partial(\partial_0 \psi_a(x))} = i\, \psi_a^*(x), \qquad \Pi_a^*(x) \equiv \frac{\partial \mathcal{L}}{\partial(\partial_0 \psi_a^*(x))} = 0. \tag{2.3}$$

Canonical quantization at $t = 0$ promotes fields to operators and imposes

$$[\hat{\psi}_a(\mathbf{x}), \hat{\Pi}_b(\mathbf{y})] = i\, \delta^{(3)}(\mathbf{x}-\mathbf{y})\, \delta_{ab}, \qquad [\hat{\psi}_a(\mathbf{x}), \hat{\psi}_b(\mathbf{y})] = [\hat{\Pi}_a(\mathbf{x}), \hat{\Pi}_b(\mathbf{y})] = 0, \tag{2.4}$$

with the rule that for fermions commutators are replaced by anticommutators.

The general solution becomes an operator expansion

$$\hat{\psi}(\mathbf{x},t) = \int \frac{d^3p}{(2\pi)^3} \sum_{\lambda=-s}^{s} \hat{a}_\lambda(\mathbf{p})\, \chi_\lambda\, e^{-iEt+i\mathbf{p}\cdot\mathbf{x}}, \qquad E = \frac{\mathbf{p}^2}{2m}, \tag{2.5}$$

with $[\hat{a}_\lambda(\mathbf{p}), \hat{a}^\dagger_{\lambda'}(\mathbf{p}')] = (2\pi)^3 \delta^{(3)}(\mathbf{p}-\mathbf{p}')\delta_{\lambda\lambda'}$ (and commutators among $\hat{a}$ or among $\hat{a}^\dagger$ vanishing, for bosons).

Assuming a vacuum state $|0\rangle$ such that $\hat{a}_\lambda(\mathbf{p})|0\rangle = 0$, one defines one-particle states $|\mathbf{p}\,\lambda\rangle = \hat{a}^\dagger_\lambda(\mathbf{p})|0\rangle$. Multi-particle states follow similarly, spanning the *Fock space*.

2.2.3 Discrete symmetries: parity and time reversal

Parity acts as $\mathbf{x} \to -\mathbf{x}$ and $\mathbf{p} \to -\mathbf{p}$, with $\mathbf{L} \to \mathbf{L}$ and $\mathbf{S} \to \mathbf{S}$. In QFT it is implemented by a unitary operator P such that $P^2 = 1$ and

$$P\, \psi(t,\mathbf{x})\, P^{-1} = \pm\psi(t,-\mathbf{x}), \qquad P\, a_\lambda(\mathbf{p})\, P^{-1} = \pm a_\lambda(-\mathbf{p}). \tag{2.6}$$

Time reversal acts as $\mathbf{x} \to \mathbf{x}$, $\mathbf{p} \to -\mathbf{p}$, $\mathbf{L} \to -\mathbf{L}$, $\mathbf{S} \to -\mathbf{S}$. It is implemented by an *antiunitary* operator T.

2.2.4 Coupling to electromagnetism: gauge invariance and minimal coupling

The Maxwell equations in vacuum follow from

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \tag{2.7}$$

which is invariant under the gauge transformation $A_\mu \to A_\mu - \partial_\mu \theta(x)$. For a charged Schrödinger field, gauge symmetry acts as $\psi(x) \to e^{iq\theta(x)}\psi(x)$. Gauge invariance is ensured by *minimal coupling*

$$\partial_\mu \to D_\mu \equiv \partial_\mu + iqA_\mu, \tag{2.8}$$

leading schematically to $\mathcal{L} = \mathcal{L}_{\text{EM}} + \psi^\dagger \left(iD_0 + \frac{\mathbf{D}^2}{2m} + \cdots \right) \psi$. If $s \neq 0$, non-minimal couplings (e.g. magnetic moment terms) may also exist.

2.3 Scalar fields: Klein–Gordon (spin 0)

2.3.1 Wave equation and Lagrangian

The simplest relativistic wave equation is the Klein–Gordon equation

$$(\partial_\mu \partial^\mu + m^2)\, \phi(x) = 0. \tag{2.9}$$

For a complex scalar field, it follows from

$$\mathcal{L} = (\partial_\mu \phi^*)(\partial^\mu \phi) - m^2 \phi^* \phi. \tag{2.10}$$

For a real scalar field, one uses

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2} m^2 \phi^2. \tag{2.11}$$

2.3.2 Mode expansion and particle/antiparticle operators

Upon quantization, the general solution can be written as

$$\hat{\phi}(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \left(e^{-ip\cdot x}\, \hat{a}(\mathbf{p}) + e^{ip\cdot x}\, \hat{b}^\dagger(\mathbf{p}) \right), \qquad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}. \tag{2.12}$$

Here $\hat{a}$ annihilates a spin-0 particle and $\hat{b}$ annihilates its antiparticle. For a real scalar field, particle and antiparticle coincide: $\hat{a} = \hat{b}$.

The vacuum $|0\rangle$ satisfies $\hat{a}(\mathbf{p})|0\rangle = \hat{b}(\mathbf{p})|0\rangle = 0$. With relativistic normalization, one often chooses factors so that $\langle \mathbf{p} \,|\, \mathbf{p}\,' \rangle = 2E_{\mathbf{p}}(2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}\,')$.

2.3.3 Discrete symmetries P, C and T

Parity can be implemented as

$$P\, \phi(t, \mathbf{x})\, P^{-1} = \pm\phi(t, -\mathbf{x}), \qquad P\, a(\mathbf{p})\, P^{-1} = \pm a(-\mathbf{p}), \quad P\, b(\mathbf{p})\, P^{-1} = \pm b(-\mathbf{p}), \tag{2.13}$$

so particle and antiparticle have the *same* intrinsic parity for scalars.

Charge conjugation exchanges particle and antiparticle operators:

$$C\,a(\mathbf{p})\,C^{-1} = b(\mathbf{p}), \qquad C\,b(\mathbf{p})\,C^{-1} = a(\mathbf{p}), \tag{2.14}$$

and acts on the field as $C\,\phi(x)\,C^{-1} = \phi^*(x)$. For a real scalar field, C can act as $C\phi C^{-1} = \pm\phi$ (defining C-parity).

Time reversal can be implemented as

$$T\,\phi(t,\mathbf{x})\,T^{-1} = \eta_T\,\phi(-t,\mathbf{x}), \qquad |\eta_T| = 1. \tag{2.15}$$

2.3.4 Coupling to electromagnetism

Assign the gauge transformation $\phi(x) \to e^{iq\theta(x)}\phi(x)$ and apply minimal coupling:

$$\partial_\mu \to D_\mu = \partial_\mu + iqA_\mu, \qquad \mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + (D_\mu\phi)^*(D^\mu\phi) - m^2\phi^*\phi. \tag{2.16}$$

Minimal coupling implies that particle and antiparticle carry opposite charges. For a real scalar field, only non-minimal couplings are allowed.

2.4 Dirac fields: spin 1/2

2.4.1 Dirac equation and Lagrangian

The relativistic wave equation for spin-1/2 particles is the Dirac equation

$$(i\gamma^\mu\partial_\mu - m)\,\psi(x) = 0, \tag{2.17}$$

where ψ is a complex 4-component spinor and the gamma matrices satisfy

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}. \tag{2.18}$$

The corresponding Lagrangian is

$$\mathcal{L} = \bar{\psi}\,(i\partial\!\!\!/ - m)\psi, \qquad \bar{\psi} \equiv \psi^\dagger\gamma^0, \qquad \partial\!\!\!/ \equiv \gamma^\mu\partial_\mu. \tag{2.19}$$

2.4.2 Mode expansion and quantization

The quantized field admits the standard expansion

$$\hat{\psi}(x) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_{\mathbf{p}}}}\sum_{\lambda=\pm}\left(e^{-ip\cdot x}\,u_\lambda(\mathbf{p})\,\hat{a}_\lambda(\mathbf{p}) + e^{ip\cdot x}\,v_\lambda(\mathbf{p})\,\hat{b}^\dagger_\lambda(\mathbf{p})\right), \qquad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}, \tag{2.20}$$

where u_λ and v_λ satisfy

$$(p\!\!\!/ - m)u_\lambda(\mathbf{p}) = 0, \qquad (p\!\!\!/ + m)v_\lambda(\mathbf{p}) = 0, \qquad p\!\!\!/ \equiv \gamma^\mu p_\mu. \tag{2.21}$$

Consistency requires *fermionic* quantization: the creation/annihilation operators satisfy anticommutation relations.

2.4.3 Gamma-matrix technology (useful identities)

A complete basis of 4×4 matrices is $\{I, \gamma^5, \gamma^\mu, \gamma^5\gamma^\mu, \sigma^{\mu\nu}\}$ with $\sigma^{\mu\nu} \equiv \frac{i}{2}[\gamma^\mu, \gamma^\nu]$. Important trace properties include:

- ◇ $\mathrm{tr}(\text{odd number of } \gamma) = 0$,
- ◇ $\mathrm{tr}(\gamma^\mu\gamma^\nu) = 4g^{\mu\nu}$,
- ◇ $\mathrm{tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho})$,
- ◇ $\mathrm{tr}(\gamma^5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4i\epsilon^{\mu\nu\rho\sigma}$.

2.4.4 Discrete symmetries P, C and T

A standard implementation is:

$$P\,\psi(t,\mathbf{x})\,P^{-1} = \gamma^0\,\psi(t,-\mathbf{x}), \tag{2.22}$$
$$C\,\psi(x)\,C^{-1} = i\gamma^2\,\psi^*(x) \equiv \psi^c(x), \tag{2.23}$$
$$T\,\psi(t,\mathbf{x})\,T^{-1} = \eta_T\,\gamma^1\gamma^3\,\psi(-t,\mathbf{x}), \tag{2.24}$$

with η_T a phase. One consequence is that particle and antiparticle have *opposite* parity for Dirac fields (visible already at the level of how a_λ and b_λ transform).

2.4.5 Coupling to electromagnetism

Gauge invariance under $\psi \to e^{iq\theta(x)}\psi$ is ensured by minimal coupling:

$$\partial_\mu \to D_\mu = \partial_\mu + iqA_\mu, \qquad \mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\not{D} - m)\psi. \tag{2.25}$$

Particle and antiparticle have opposite electric charge.

2.5 Chirality, helicity and Weyl fermions

Define the chiral projectors

$$P_R = \frac{1+\gamma^5}{2}, \qquad P_L = \frac{1-\gamma^5}{2}, \qquad \psi_R \equiv P_R\psi, \qquad \psi_L \equiv P_L\psi, \tag{2.26}$$

with $P_R + P_L = 1$, $P_R^2 = P_R$, $P_L^2 = P_L$, and $P_RP_L = 0$.

In the massless limit (equivalently, at very high energies $E \gg m$), right- and left-handed components decouple:

$$\bar{\psi}(i\not{\partial} - m)\psi = \bar{\psi}_R i\not{\partial}\psi_R + \bar{\psi}_L i\not{\partial}\psi_L - m\bar{\psi}_R\psi_L - m\bar{\psi}_L\psi_R \xrightarrow[E\gg m]{} \bar{\psi}_R i\not{\partial}\psi_R + \bar{\psi}_L i\not{\partial}\psi_L. \tag{2.27}$$

In this regime, chirality aligns with helicity: right-handed fields describe particles with positive helicity (and antiparticles with negative helicity), while left-handed fields describe particles with negative helicity (and antiparticles with positive helicity). The two-component Weyl equations arise as the building blocks of the electroweak theory.

Discrete symmetries act non-trivially on chirality; in particular, parity exchanges left and right components.

2.6 Majorana masses and Majorana fermions

Besides the usual Dirac mass term, one can add a *Majorana mass* term of the schematic form

$$\delta\mathcal{L} = -\delta m\,(\bar{\psi}\,\psi^c + \bar{\psi}^c\,\psi), \tag{2.28}$$

which violates the global U(1) symmetry $\psi \to e^{i\theta}\psi$ and therefore does not conserve "particle number minus antiparticle number".

One can define a Majorana fermion by the condition

$$\psi = \psi^c = i\gamma^2\psi^*, \tag{2.29}$$

implying that particle and antiparticle operators coincide (roughly $a_\lambda = b_\lambda$), so ψ and ψ^* are no longer independent.

2.7 Vector fields: Proca and Maxwell (spin 1)

2.7.1 Equation of motion and Lagrangians

A massive spin-1 field B_μ satisfies the Proca equation

$$\partial_\mu B^{\mu\nu} + m^2 B^\nu = 0, \qquad B^{\mu\nu} = \partial^\mu B^\nu - \partial^\nu B^\mu. \tag{2.30}$$

For $m = 0$ this reduces to Maxwell's equations (in vacuum) for a gauge field.

For a *real* vector field the Lagrangian is

$$\mathcal{L} = -\frac{1}{4}B_{\mu\nu}B^{\mu\nu} + \frac{1}{2}m^2 B_\mu B^\mu. \tag{2.31}$$

For a *complex* vector field,

$$\mathcal{L} = -\frac{1}{2}B^*_{\mu\nu}B^{\mu\nu} + m^2 B^*_\mu B^\mu. \tag{2.32}$$

2.7.2 Mode expansion and polarizations

For $m \neq 0$, the quantized field can be expanded as

$$B_\mu(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \sum_{h=0,\pm} \left(e^{-ip\cdot x}\, \epsilon_\mu^{(h)}(\mathbf{p})\, a_h(\mathbf{p}) + e^{ip\cdot x}\, \epsilon_\mu^{(h)*}(\mathbf{p})\, a_h^\dagger(\mathbf{p}) \right), \tag{2.33}$$

with transversality $p^\mu \epsilon_\mu^{(h)}(\mathbf{p}) = 0$ and completeness relations that project onto the physical spin-1 subspace.

2.7.3 Discrete symmetries and coupling to electromagnetism

Parity, charge conjugation and time reversal can be implemented consistently; for complex fields, charge conjugation exchanges particle and antiparticle operators $a_h \leftrightarrow b_h$.

If B_μ is complex and charged, gauge invariance under $B_\mu \to e^{iq\theta(x)} B_\mu$ is ensured by minimal coupling:

$$\partial_\mu \to D_\mu = \partial_\mu + iqA_\mu, \qquad \tilde{B}_{\mu\nu} \equiv D_\mu B_\nu - D_\nu B_\mu, \tag{2.34}$$

leading to a gauge-invariant Lagrangian of the form

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} \tilde{B}^*_{\mu\nu} \tilde{B}^{\mu\nu} + m^2 B^*_\mu B^\mu. \tag{2.35}$$

For a real vector field, only non-minimal couplings are allowed (unless the field itself is the gauge field).

2.8 Summary

The free-field construction illustrates the general QFT pattern:

- ◇ Start from a wave equation, derive it from a Lagrangian,
- ◇ quantize canonically to obtain creation/annihilation operators and Fock space,
- ◇ implement discrete symmetries via unitary/antiunitary operators,
- ◇ include interactions by adding local terms that respect the symmetries (e.g. gauge invariance via minimal coupling).

This framework will be the foundation for interacting theories and for computing physical observables in the SM.

3 . The Quark Model and Effective Theories of Hadrons

3.1 Motivation: why do we need more than the quark model?

So far we have used approximate internal symmetries (isospin and flavor SU(3)) to organize hadrons into multiplets. Symmetry, however, does not by itself provide the *dynamics*: it does not explain *why* some hadrons are much lighter than others, nor does it systematically predict low-energy interactions.

A first dynamical attempt is the (constituent) non-relativistic quark model: hadrons are bound states of quasi-particles (constituent quarks) moving slowly, with masses approximately given by sums of constituent masses plus small interaction corrections. This picture is qualitatively useful for spectroscopy, but it becomes internally inconsistent in the light-hadron sector (especially because of the pion). The goal of this chapter is:

- ◇ to see *why* the non-relativistic quark model fails for light hadrons,
- ◇ to understand the pion as a (pseudo-)Goldstone boson of chiral symmetry breaking,
- ◇ to introduce the EFT logic (chiral perturbation theory) describing low-energy hadron physics.

For broader context on the quark model and hadron spectroscopy, see standard particle-physics texts such as (Halzen and Martin, 1984; Griffiths, 2008; Thomson, 2013), and for QFT background and symmetry concepts see (Peskin and Schroeder, 1995; Schwartz, 2014; Weinberg, 1995).

3.2 The non-relativistic quark model

3.2.1 Basic assumption and hyperfine interaction

The simplest dynamical assumption is:

Quarks move slowly inside hadrons, so hadron masses are roughly sums of constituent quark masses, plus relatively small interaction corrections.

To account for splittings between spin multiplets (e.g. 0^- vs. 1^- mesons and $\frac{1}{2}^+$ vs. $\frac{3}{2}^+$

baryons), the effective interaction must depend on spin. A common schematic ingredient is a hyperfine (spin–spin) Hamiltonian of the form

$$H_{\rm hf} = a \sum_{i \neq j} \frac{\mathbf{S}_i \cdot \mathbf{S}_j}{m_i m_j}, \tag{3.1}$$

where m_i are constituent masses, $\mathbf{S}_i$ are spin operators, and a is an effective constant encoding interaction details and wavefunction overlap. This structure is widely used in introductory discussions of hadron spectroscopy (Halzen and Martin, 1984; Griffiths, 2008; Thomson, 2013).

3.2.2 Mesons: $q_1\bar{q}_2$

For mesons, a schematic mass formula is

$$M_{q_1\bar{q}_2} \;=\; m_1 + m_2 + \frac{a}{m_1 m_2}\left(\frac{S(S+1)}{2} - \frac{3}{2}\right), \qquad S = \text{meson spin}, \tag{3.2}$$

where the spin factor comes from $\langle \mathbf{S}_1 \cdot \mathbf{S}_2 \rangle$. In the light SU(2)/SU(3) limit one may take $m_1 = m_2 = m$, giving

$$J^P = 0^- : \quad M = 2m - \frac{3a}{4m^2}, \tag{3.3}$$

$$J^P = 1^- : \quad M' = 2m + \frac{a}{4m^2}. \tag{3.4}$$

If we (naively) identify these with π and ρ, then

$$M' - M \;=\; \frac{a}{m^2} \;\simeq\; m_\rho - m_\pi \;\approx\; 630 \text{ MeV}. \tag{3.5}$$

In such a fit the hyperfine splitting scale is not "small" compared to typical constituent masses (one often finds m of order a few hundred MeV). This already signals a breakdown of the underlying non-relativistic hypothesis in the light sector: the "interaction correction" is comparable to (or larger than) the would-be constituent mass scale. The core issue is not the specific numbers, but the *self-consistency* of the expansion assumed by the model (Griffiths, 2008; Thomson, 2013).

3.2.3 Baryons: $q_1 q_2 q_3$

For baryons a similar schematic expression is

$$M_{q_1 q_2 q_3} \;=\; m_1 + m_2 + m_3 + a' \sum_{i \neq j} \frac{\langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle}{m_i m_j}. \tag{3.6}$$

In the simplified SU(3)-symmetric limit $m_1 = m_2 = m_3 = m$, the spin dependence can be written as

$$M_{qqq} \;=\; 3m + \frac{a'}{2m^2}\left(S(S+1) - \frac{9}{4}\right), \qquad S = \text{baryon spin}. \tag{3.7}$$

For $J^P = \frac{1}{2}^+$ (nucleon) and $J^P = \frac{3}{2}^+$ (Δ),

$$J^P = \frac{1}{2}^+ : \quad M = 3m - \frac{3a'}{4m^2}, \tag{3.8}$$

$$J^P = \frac{3}{2}^+ : \quad M' = 3m + \frac{3a'}{4m^2}. \tag{3.9}$$

Hence

$$M' - M = \frac{3a'}{2m^2}. \tag{3.10}$$

Numerically, the N–Δ splitting is a few hundred MeV, again not parametrically small relative to typical constituent scales. The model can be a useful organizer of spectra, but it does not provide a controlled low-energy expansion for light hadrons (Halzen and Martin, 1984; Griffiths, 2008; Thomson, 2013).

3.2.4 Conclusion: inconsistency for light hadrons

The central message is:

- ◇ The non-relativistic quark model assumes a small interaction correction to constituent masses.
- ◇ In the light sector (where the pion is exceptionally light), the fitted "hyperfine" effects are not small compared to the masses being corrected.

Therefore, for low-energy QCD we need a different viewpoint. The correct language is: *symmetries plus effective field theory*. In particular, chiral symmetry and its breaking control the pion sector and organize interactions as an expansion in momenta and quark masses (Peskin and Schroeder, 1995; Weinberg, 1995; Schwartz, 2014; Donoghue et al., 1992).

3.3 The linear sigma model and chiral symmetry breaking

3.3.1 The pion puzzle

Experimentally,

$$m_\pi \sim 140 \text{ MeV} \ll m_\rho \sim 770 \text{ MeV}. \tag{3.11}$$

This suggests that the up and down quark masses are small compared to the QCD scale. In the limit $m_u \simeq m_d \simeq 0$:

- ◇ left- and right-handed Dirac fields decouple,
- ◇ isospin SU(2) enlarges to the **chiral symmetry**

$$SU_L(2) \otimes SU_R(2). \tag{3.12}$$

We do *not* observe approximate parity doublets throughout the hadron spectrum, so chiral symmetry cannot be realized linearly on the spectrum in an unbroken way. The

standard interpretation is that it is **spontaneously broken** by the vacuum, producing (nearly) massless Goldstone bosons: the pions (Peskin and Schroeder, 1995; Weinberg, 1995; Schwartz, 2014; Donoghue et al., 1992).

3.3.2 Mesonic realization

A convenient mesonic field realizing $SU_L(2) \otimes SU_R(2)$ is a 2×2 matrix

$$M(x) = \sigma(x) I_2 + i\,\tau \cdot \phi(x), \qquad \sigma, \phi \in \mathbb{R}, \tag{3.13}$$

where ϕ are pseudoscalars (identified with pions) and σ is a scalar. Consider the Lagrangian

$$\mathcal{L} = \frac{1}{4}\,\mathrm{tr}\big(\partial_\mu M^\dagger \partial^\mu M\big) - \frac{m^2}{4}\,\mathrm{tr}\big(M^\dagger M\big) - \lambda\left[\mathrm{tr}\big(M^\dagger M\big)\right]^2, \qquad \lambda > 0, \tag{3.14}$$

which is invariant under

$$M \to g_L\, M\, g_R^\dagger, \qquad g_L \in SU_L(2), \quad g_R \in SU_R(2), \tag{3.15}$$

with isospin SU(2) corresponding to $g_L = g_R$. This is the textbook linear sigma model realization of chiral symmetry (Peskin and Schroeder, 1995; Schwartz, 2014; Weinberg, 1995).

3.3.3 Vacuum structure and Goldstone bosons

Define the invariant

$$\rho \equiv \frac{1}{2}\,\mathrm{tr}(M^\dagger M), \tag{3.16}$$

so for constant fields the potential is

$$V(\rho) = \frac{m^2}{2}\rho + 4\lambda\rho^2. \tag{3.17}$$

If $m^2 > 0$, the minimum is at $\rho = 0$ and scalars/pseudoscalars are degenerate. If $m^2 < 0$, the minimum is at

$$0 = \frac{m^2}{2} + 8\lambda\rho \qquad \Rightarrow \qquad \rho_0 = -\frac{m^2}{16\lambda} > 0. \tag{3.18}$$

Isospin invariance selects a vacuum proportional to the identity:

$$M_0 = \sqrt{\rho_0}\, I_2 \equiv f_\pi\, I_2. \tag{3.19}$$

A convenient parameterization of the vacuum manifold is

$$M(x) = \big(f_\pi + S(x)\big)\, U(x), \qquad U(x) \in SU(2), \quad S(x) \in \mathbb{R}, \tag{3.20}$$

where $U(x)$ contains the Goldstone modes. If we focus on the Goldstone sector (set $S = 0$

at leading order), the kinetic term becomes

$$\mathcal{L}_{\text{Goldstone}} = \frac{f_\pi^2}{4}\,\mathrm{tr}\big(\partial_\mu U^\dagger \partial^\mu U\big)\,, \tag{3.21}$$

which is the non-linear sigma model.

Writing

$$U(x) = \exp\left(\frac{i\,\pi(x)\cdot\tau}{f_\pi}\right), \qquad \pi\cdot\tau = \begin{pmatrix} \pi^0 & \sqrt{2}\,\pi^+ \\ \sqrt{2}\,\pi^- & -\pi^0 \end{pmatrix}, \qquad \pi^\pm = \frac{\pi^1 \mp i\pi^2}{\sqrt{2}}, \tag{3.22}$$

and expanding shows that the π^a are massless when chiral symmetry is exact. This is the Goldstone pattern

$$SU_L(2)\otimes SU_R(2) \;\longrightarrow\; SU(2) \qquad \Rightarrow \qquad 3 \text{ Goldstone bosons.} \tag{3.23}$$

For the general QFT mechanism of spontaneous symmetry breaking and Goldstone bosons, see (Peskin and Schroeder, 1995; Schwartz, 2014; Weinberg, 1995).

3.3.4 The scalar mode and explicit breaking

In (3.20), the real field $S(x)$ is the heavy scalar mode. From the potential (for $m^2 < 0$) one finds a nonzero scalar mass scale, schematically

$$m_S^2 \sim -2m^2 > 0, \tag{3.24}$$

so S can be integrated out at sufficiently low energies.

Small quark masses $m_u \simeq m_d \equiv m_q$ explicitly break chiral symmetry while preserving isospin. At the level of the effective mesonic theory, one can represent explicit breaking through a term

$$\delta\mathcal{L} = \frac{f_\pi^2 B}{2}\,\mathrm{tr}\big(M_q\,U^\dagger + U\,M_q^\dagger\big)\,, \qquad M_q = m_q I_2, \tag{3.25}$$

where B is a low-energy constant. Expanding $U = \exp(i\pi^a\tau^a/f_\pi)$ to quadratic order gives the parametric scaling

$$m_\pi^2 \;\propto\; m_q, \qquad \Rightarrow \qquad m_\pi \;\propto\; \sqrt{m_q}, \tag{3.26}$$

so pions are light because they are pseudo-Goldstone bosons. This is one of the key low-energy lessons encoded systematically in chiral EFT (Donoghue et al., 1992; Schwartz, 2014; Thomson, 2013).

3.4 Including nucleons: why m_N does not vanish as $m_q \to 0$

Let $\psi = (p, n)^T$ be the nucleon isodoublet, with chiral projections $\psi_{L,R}$. A direct nucleon mass term $\bar{\psi}\psi$ violates chiral symmetry. However, a chiral-invariant Yukawa structure is

$$\mathcal{L}_N = \bar{\psi}_L i\partial\!\!\!/ \psi_L + \bar{\psi}_R i\partial\!\!\!/ \psi_R - d\,\bar{\psi}_L M \psi_R - d\,\bar{\psi}_R M^\dagger \psi_L. \tag{3.27}$$

When M condenses, $M \to f_\pi I_2$, this generates

$$m_N = d\, f_\pi, \tag{3.28}$$

which remains finite in the chiral limit. Therefore,

$$m_\pi \to 0 \;\; (m_q \to 0), \qquad \text{but} \qquad m_N \sim f_\pi \neq 0. \tag{3.29}$$

Physically, most of the nucleon mass is tied to the scale of spontaneous chiral symmetry breaking, not directly to the small explicit quark masses (Peskin and Schroeder, 1995; Schwartz, 2014; Donoghue et al., 1992).

At low energies, explicit breaking induces corrections analytic in m_q, so one expects at leading order

$$m_N = d f_\pi + \mathcal{O}(m_q), \tag{3.30}$$

while the pion mass begins already at order m_q as in (3.26).

3.5 The non-linear sigma model and chiral perturbation theory (EFT)

3.5.1 Low-energy regime and building blocks

At momenta well below the hadronic scale, and in particular for processes with

$$p \ll m_N, \tag{3.31}$$

pions can be produced, but most other hadrons cannot. This motivates an EFT built only out of pion degrees of freedom: **chiral perturbation theory**.

The building blocks are:

- the pion field matrix $U(x) \in SU(2)$, with $U^\dagger U = 1$,
- an explicit-breaking spurion M_q that transforms as $M_q \to g_L M_q g_R^\dagger$ and is set at the end to $M_q = m_q I_2$.

Power counting is

$$\partial_\mu \sim p \sim m_\pi, \qquad M_q \sim m_q \sim m_\pi^2 \sim p^2. \tag{3.32}$$

Then one writes *all* operators consistent with Lorentz symmetry, chiral symmetry (with spurions), parity, and charge conjugation, organized by order in p. The UV physics is

encoded in low-energy constants multiplying these operators. This is the general EFT philosophy in a concrete hadronic setting (Donoghue et al., 1992; Schwartz, 2014; Weinberg, 1995).

3.5.2 Leading order: only two universal terms

At leading order, $\mathcal{O}(p^2)$, the pion EFT has the universal form

$$\mathcal{L}_\chi^{(2)} = \frac{f_\pi^2}{4}\,\mathrm{tr}\big(\partial_\mu U^\dagger \partial^\mu U\big) + \frac{f_\pi^2 B}{2}\,\mathrm{tr}\big(M_q\,U^\dagger + U\,M_q^\dagger\big)\,. \tag{3.33}$$

Thus, at very low energies pion physics depends essentially on f_π and on one additional constant B (which sets m_π once M_q is fixed). At next-to-leading order, $\mathcal{O}(p^4)$, many additional operators appear, each with its own low-energy constant, encoding short-distance QCD effects (Donoghue et al., 1992; Schwartz, 2014).

3.6 Pion–nucleon EFT and the axial coupling g_A

3.6.1 Chiral-covariant building blocks

To include nucleons while keeping a controlled expansion for $p \ll m_N$, it is convenient to introduce $u(x)$ such that

$$U(x) = u(x)^2. \tag{3.34}$$

Define

$$v_\mu \equiv \frac{1}{2}\left(u\partial_\mu u^\dagger + u^\dagger \partial_\mu u\right), \qquad a_\mu \equiv \frac{1}{2}\left(u\partial_\mu u^\dagger - u^\dagger \partial_\mu u\right), \tag{3.35}$$

which transform as chiral-covariant vector/axial combinations. Then the leading pion–nucleon Lagrangian can be written schematically as

$$\mathcal{L}_{\pi N}^{(1)} = \bar{N}\left(i\partial\!\!\!/ + v\!\!\!/\right)N + g_A\,\bar{N}\,a\!\!\!/\gamma^5 N - m_N\,\bar{N}N + \cdots\,, \tag{3.36}$$

where the ellipsis denotes higher-order terms in the derivative/mass expansion. The coefficient multiplying the axial structure is the **axial coupling** g_A. It is not fixed by symmetry alone; it must be taken from experiment. This is a typical EFT feature: symmetry fixes the operator basis and power counting, but not the numerical values of the low-energy constants (Donoghue et al., 1992; Schwartz, 2014; Thomson, 2013).

3.6.2 Very low-energy scattering

Expanding u in pion fields yields, schematically,

$$a_\mu \simeq \frac{i}{2f_\pi}\,\partial_\mu(\pi\cdot\tau) + \cdots\,, \qquad v_\mu \simeq \mathcal{O}\left(\frac{\pi\,\partial\pi}{f_\pi^2}\right), \tag{3.37}$$

so at the lowest energies pion–nucleon interactions are governed primarily by f_π and g_A through the leading terms in (3.36). This explains why a small set of measured constants

can control a large class of low-energy processes in the pion sector (Donoghue et al., 1992; Schwartz, 2014).

3.7 Final remarks and SU(3) generalization

- ◇ The heavy scalar mode in the linear sigma model is *model-dependent*. The *universal* low-energy content is encoded in the EFT for $U(x)$ (and in the pion–nucleon EFT if baryons are included).
- ◇ The general EFT logic (separation of scales, symmetry-based operator expansion, and low-energy constants) is part of the standard QFT toolkit (Peskin and Schroeder, 1995; Schwartz, 2014; Weinberg, 1995).

Finally, the non-linear sigma model generalizes to flavor SU(3). In the idealized limit $m_u \simeq m_d \simeq m_s \simeq 0$ one has

$$SU_L(3) \otimes SU_R(3) \longrightarrow SU(3), \tag{3.38}$$

with $U(x) \in SU(3)$ written as

$$U(x) = \exp\left(\frac{i\,\lambda_a \phi^a}{f}\right), \tag{3.39}$$

so that kaons and the η appear as additional approximate Goldstone bosons. In the isospin limit, explicit breaking can be represented by a mass spurion

$$M_q = \begin{pmatrix} m_q & 0 & 0 \\ 0 & m_q & 0 \\ 0 & 0 & m_s \end{pmatrix}, \tag{3.40}$$

which generates the observed mass pattern among the pseudo-Goldstone multiplet (Thomson, 2013; Donoghue et al., 1992).

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